Lecture 33: Topological Consideration in Maxwell's Equations

Curl: Recall $\vec{v}$ is a velocity field.

$$\text{curl } \vec{v}$$

measures twice the angular velocity vector (rotation part of motion).

Example: Uniform Rotation about the z-axis

$$\vec{v} = \omega(-y\hat{i} + x\hat{j})$$

$$\nabla \times \vec{v} = 2\omega\hat{k}$$

For a force field:
Torque:

$$\vec{\tau} = \vec{r} \times \vec{F}\,\Delta m$$

for a mass element Δm.
For translation:

$$\text{Force} = \text{mass} \times \text{acceleration}$$

$$F = m\frac{d(\text{velocity})}{dt}$$

For rotation:

$$\text{Torque} = (\text{moment of inertia}) \times (\text{angular acceleration})$$

$$\tau = I\frac{d(\text{angular velocity})}{dt}$$

Consequence: If force $\vec{F}$ derives ...
From a potential,

$$\nabla \times \vec{F} = 0$$

$$\vec{F} \text{ does not induce any rotational motion}$$

Electric and Magnetic Fields

Maxwell's equations

$$\vec{F} = q\vec{E}$$

$$\vec{F} = q\vec{v} \times \vec{B}$$

1. Divergence and Curl

Gauss Coulomb Law

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

where

$$\varepsilon_0 = \text{constant}$$

and ρ is the electric charge density.

$$\begin{aligned} \iint_S \vec{E} \cdot d\vec{S} &= \iiint_V (\nabla \cdot \vec{E})\, dV = \frac{1}{\varepsilon_0} \iiint_V \rho \, dV \\ &= \frac{Q}{\varepsilon_0} \end{aligned}$$

where Q is the electric charge enclosed.

Faraday's Law

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

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Stokes' Theorem

$$\oint_C \vec{E} \cdot d\vec{r} = \iint_S (\nabla \times \vec{E}) \cdot d\vec{S} = \iint_S \left(-\frac{\partial \vec{B}}{\partial t} \right) \cdot d\vec{S}$$

Remarks

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \varepsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

where $\vec{J}$ is the vector current density.